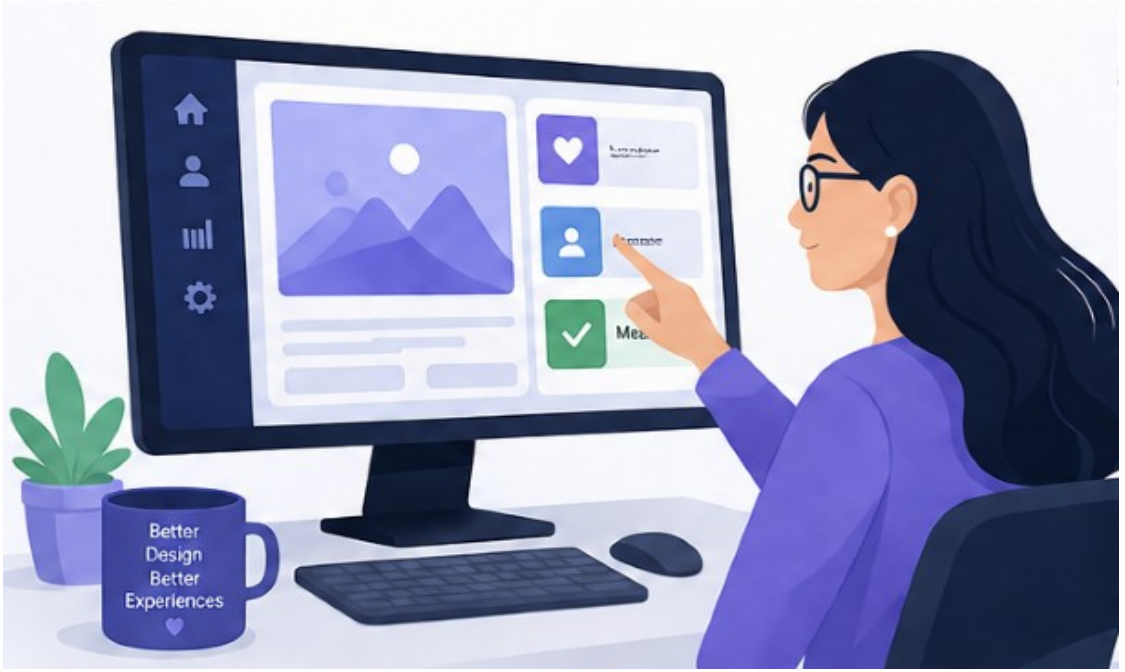
HUMAN COMPUTER INTERACTION

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HUMAN COMPUTER INTERACTION (HCI)

- Course Learning Outcomes: Upon successful completion of the course, you will be able to:

CLOs	Description	Taxonomy level	PLO
1	**Identify various interaction mechanisms and purpose of a variety of interfaces along with their application areas.**	**C2**	**1**
2	**Distinguish modules that cover the entire user-centered design and evaluation process - from understanding user needs, to designing interactive systems that meet those needs.**	**C4**	**2**

IMPORTANT TO REMEMBER

- Only theory, no practical (The teaching will be based on Examples and Case studies)
- 3 Credit Hour (100 marks)
- **Class performance matters the most !**

RECOMMENDED READING

- Human Computer Interaction Second Edition Dix, Finlay, Abowd & Beale
- User Interface Design and Evaluation Amazon by Debbie Stone, Caroline Jarret, Mark Wooroffe and Shailey Minocha
- Design Interfaces: Patterns of Effective Interaction Design by Jennifer Tidwell
- Internet AI tools.

WHAT IS HUMAN COMPUTER INTERACTION?

- HCI (Human-computer interaction) is a multidisciplinary field that focuses on optimizing how users and computers interact by designing interactive computer interfaces that satisfy users' needs.

- Multidisciplinary field.
- includes: behavioral sciences, cognitive science, human psychology, and design principles.

HUMAN COMPUTER INTERACTION

Human-Computer Interaction (HCI) is the study of how people interact with computers and how to make those interactions easier and more effective. It combines ideas from different fields like computer science, psychology, and design to create user-friendly systems.

HCI aims to make technology easy, natural, and accessible for everyone. It combines computer science, psychology, and design to ensure machines work for humans—not the other way around.

HUMAN COMPUTER INTERACTION

HCI started in the 1980s when personal computers became common in homes and offices. Over time, as technologies like the internet, mobile devices, and voice-based systems developed, HCI grew to focus on making these technologies simple and enjoyable to use.

Today, HCI is all about designing interfaces that give users a better experience.

EXAMPLE: HUMAN COGNITIVE INTERACTION

The brain is the central processing unit of the human side.
When using a computer:
1.User receives information through senses.
2.Brain interprets the information.
3.User performs an action.
4.Computer responds.
5.The cycle repeats.

Example

While using Microsoft Word:

- You **see** text on the screen.
- Your brain understands what is displayed.
- You **touch** the keyboard.
- The computer updates the document.
- You again **see** the updated result.

WHY IS HCI IMPORTANT?

The ultimate objective of HCI is to bridge the gap between human needs and machine capabilities.

By studying how we interact with technology, researchers can create user-friendly and highly accessible devices.

Good HCI translates to better user experiences (UX), higher productivity, and technology that adapts to the user rather than forcing the user to adapt to rigid, complex systems.

EXAMPLES: 1. INTERACTIVE MUSEUM EXHIBITS

Museums have evolved from static displays to highly interactive learning environments.

HCI Technologies

Multi-touch displays

Gesture recognition

Object recognition

RFID-enabled exhibits

AI-based tour guides

Interactive projection walls







Example Interaction

A visitor touches an ancient artifact displayed on a touchscreen. The system displays:

3D rotating model

Historical timeline

Audio narration

Related artifacts

Interactive quizzes

HCI Concepts

Direct manipulation

Immediate feedback

Learnability

Information visualization

User engagement







Class Exercise:

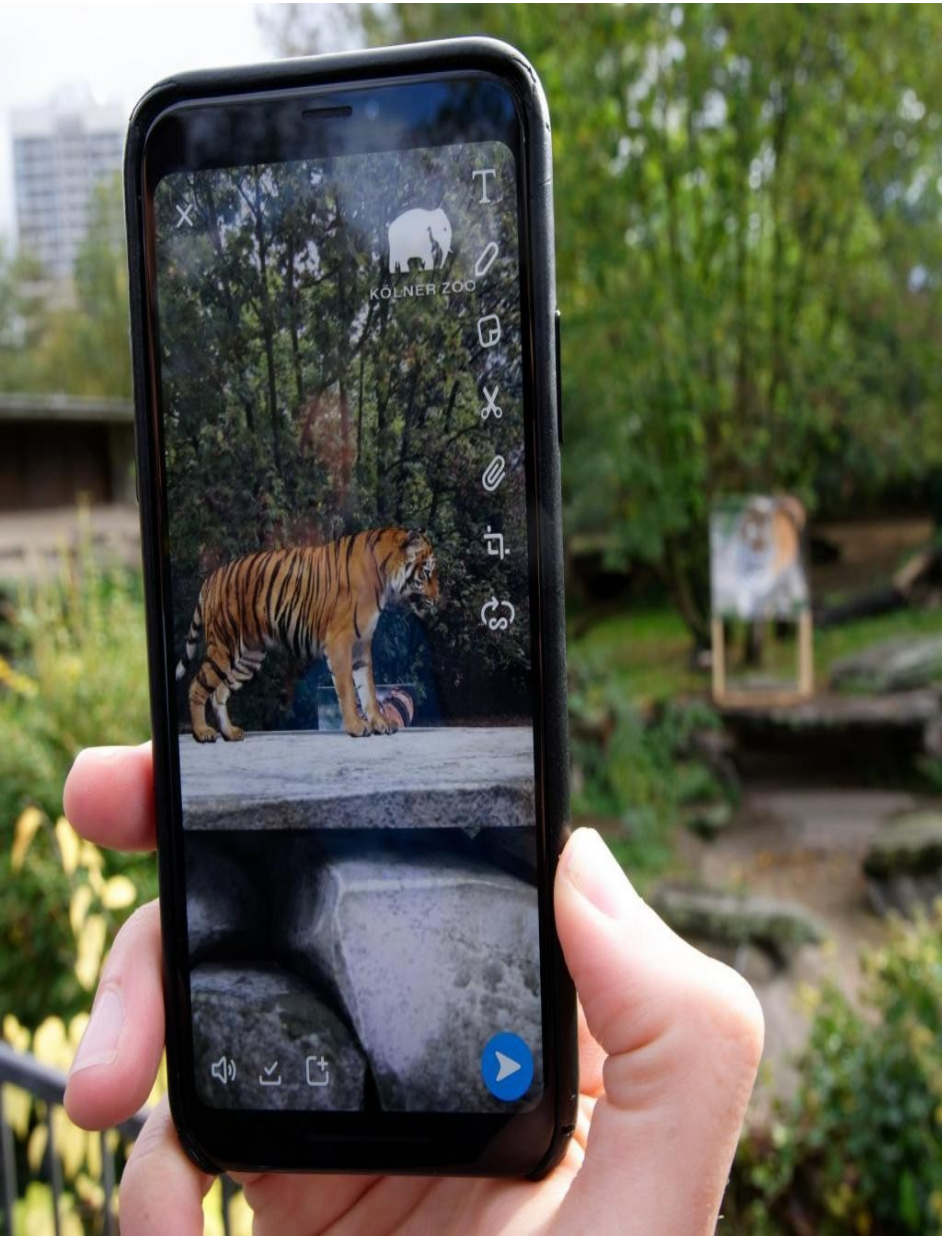
How can this exhibit be redesigned for visually impaired visitors?







2. Augmented Reality (AR) Zoo Guide





Visitors point their phone toward an animal enclosure.

The interface displays:

Animal name

Habitat (natural environment where an organism lives, grows, and reproduces)

Animated 3D model

Audio pronunciation

Detailed information

HCI Concepts

Context-aware computing (systems that sense their physical and digital environments to adapt their behavior automatically)

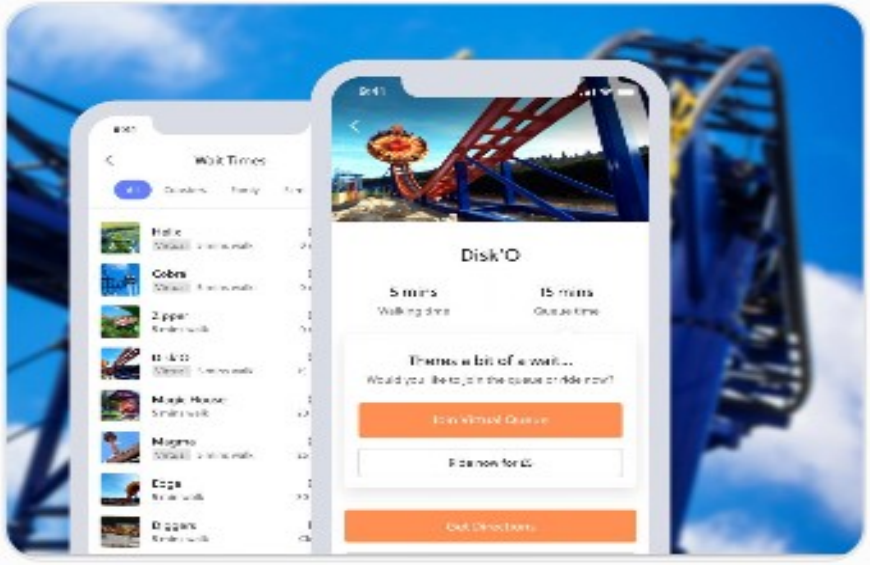
Mobile interaction

Multimodal interfaces (allows you to interact with systems using two or more distinct human communication channels such as speech, touch, gesture, and gaze simultaneously or interchangeably). Multimodal interfaces feel more natural.

Location awareness

3. SMART THEME PARKS (E.G., DISNEY-STYLE ATTRACTIONS)





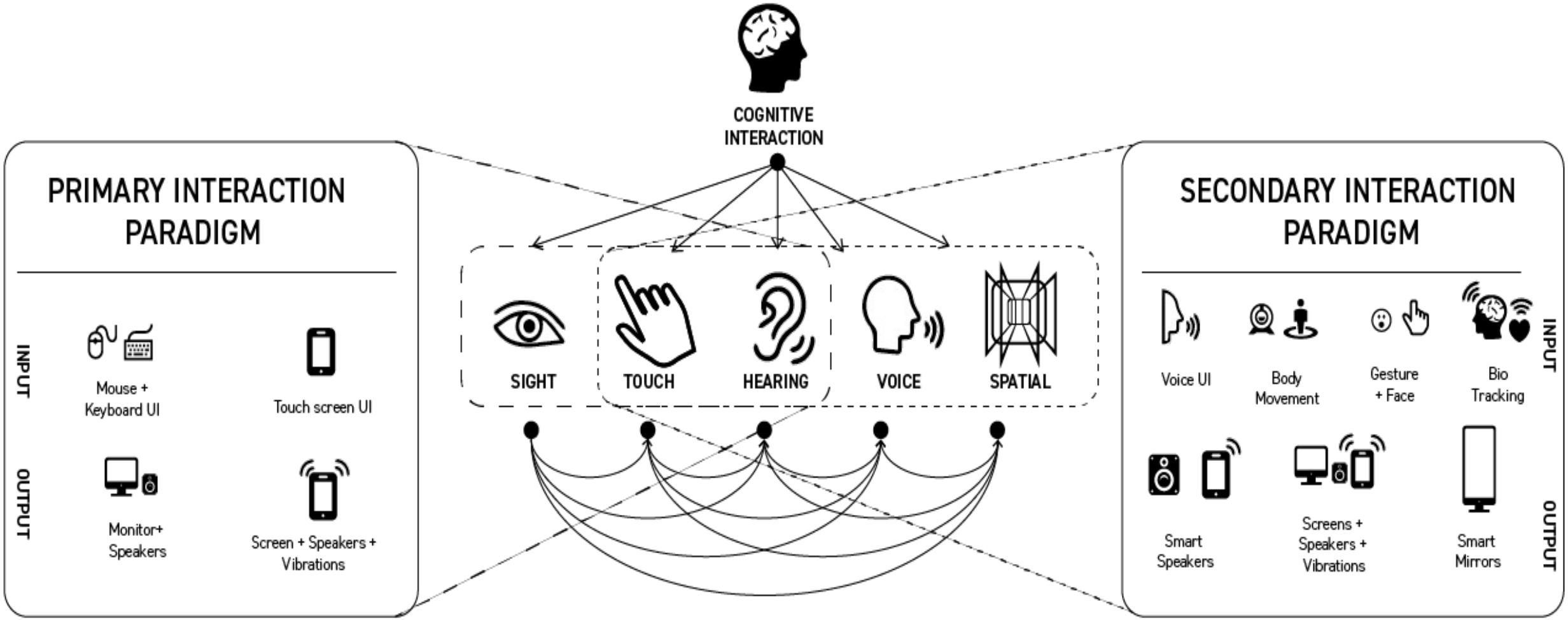


Visitors wear RFID wristbands.

The wristband enables:

- Entry
- Ride reservations
- Cashless payment
- Personalized recommendations
- Character interactions

HUMAN INTERACTION WITH A COMPUTER



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COMMON INTERACTION CHANNELS

1. **Sight/Vision (Visual Channel)**: This is the most heavily used channel in HCI.

Examples:

1. **Reading text on a screen (menus, dialog boxes, error messages)**
2. **Interpreting icons, color-coded alerts (red for danger, green for success)**
3. **Watching progress bars, animations, or data visualizations (charts, dashboards)**
4. **Reading a GPS map while navigating.**
5. **Eye Tracking: The computer watches where your eyes look. You can select items, type, or scroll just by looking at specific spots on the screen.**
6. **Gaze Gestures: You can move your eyes in a pattern (like blinking or looking left to right) to send a command to the device.**

COMMON INTERACTION CHANNELS

2.Touch (Haptic/Tactile Channel) : Refer to the human sense of touch. They are used in technology to create physical sensations

Physical contact used for both input (manipulating controls) and output (receiving feedback through vibration, texture, or resistance).

Examples:

Typing on a physical or touchscreen keyboard

Pinch-to-zoom, swipe, and tap gestures on smartphones

Phone vibration alerts for notifications or incoming calls

Force-feedback joysticks in gaming or flight simulators

VR Gloves: Special gloves physically resist your fingers. This lets you grab, hold, and "feel" the shape of a digital object. This lets you reach out, grab, and touch virtual objects naturally.

COMMON INTERACTION CHANNELS

3. Hearing (Auditory Channel): Sound-based output that conveys information, feedback, or alerts. Visually disabled people can be guided by the use of sound.

Examples:

Notification sounds (email "ping," message alert)

Voice assistants speaking responses (Siri, Alexa reading out the weather)

Camera shutter sound confirming a photo was taken

Alarm sounds in a car when a seatbelt isn't fastened

Audio cues in games (footsteps, gunfire direction)

Navigation Systems.

Screen Readers: Software used by visually impaired users to read text on a screen aloud

Earcons

COMMON INTERACTION CHANNELS

4. Voice (Speech Channel): Spoken language used as input to control or communicate with a system.

Examples:

Voice commands: "Hey Siri, set a timer for 10 minutes"

Dictating text messages or emails

Voice-controlled smart home devices ("turn off the lights")

Call center IVR systems ("Press or say 'billing'")

Voice authentication for banking apps

COMMON INTERACTION CHANNELS

5. Spatial Interaction (Movement/Gesture/Body Channel)

Interaction through body position, motion, gestures, or physical location in space; increasingly relevant with AR/VR and motion-sensing devices.

Examples:

1. Waving a hand to control a Kinect-based game (A Kinect-based game is a video game you play using your body instead of a controller. The Kinect is a camera device that tracks your movements. As you move your hands, feet, or body, the game sees you and reacts to your actions.)
2. VR headset tracking head movement to change the viewpoint
3. Tilting a phone to steer in a racing game (uses an internal sensor called a ***gyroscope*** to measure the angle of your device.)
4. Walking through an AR navigation app that overlays directions on the real world

Smartwatch detecting a "raise wrist" gesture to wake the screen

INTERACTION CHANNELS: CASE STUDY

Designing a Smart Hospital Room for Patient Monitoring

A hospital wants to design a **smart patient room system** to improve safety and comfort for post-surgery patients, especially elderly patients with limited mobility and some visual or hearing impairment. The system must let patients call for help, monitor vital signs, control room settings (lights, bed angle, TV), and alert nurses in emergencies; all while minimizing physical strain on the patient.

Constraints:

Patients may have limited hand strength or tremors (affects touch precision)

Some patients have mild hearing loss

Some patients have low vision, especially at night

Patients are often lying down, with limited range of motion (affects gesture/spatial input)

Nurses need rapid, unambiguous alerts even when not in the room

INTERACTION CHANNELS: CASE STUDY

Task

For each of the five interaction channels, students must:

Propose **one specific feature** of the smart room that uses that channel (as input, output, or both)

Identify a **potential failure point** for that channel given the patient constraints above

Suggest a **redundant or backup channel** to cover that failure (this is the key HCI lessoni .e multimodal redundancy for accessibility and safety)

Example of expected output (sight Channel, partially worked, to show format):

Feature: Large on-screen icons showing vital signs trends, color-coded for normal/warning/critical

Failure point: Patient has low vision at night or is asleep and won't see a visual alert

Backup channel: Pair visual alert with an audible chime (hearing) and a bed vibration pulse (touch) for critical alerts.

Components of Human-Computer Interaction



Users



Tasks



Tools



Interfaces



Context

KEY COMPONENTS OF HCI

1. Users

Users are central to HCI design. Researchers spend significant time understanding user needs and expectations through user research, which helps create user-centered designs. These designs prioritize usability and usefulness, ensuring the technology improves users' lives.

2. Tasks

In addition to understanding users, designers must consider the tasks users will perform with the system. Task analysis helps map out the user journey, creating scenarios to ensure the interface supports task-oriented, efficient interaction.

KEY COMPONENTS OF HCI

3. Tools

Tools are essential for gathering data, conducting analysis, prototyping, and evaluating HCI designs. UI/UX tools enable iterative design processes, helping designers refine and optimize interfaces based on user feedback and testing. Examples are Figma, Adobe XD, Pencil Project, Canva etc.

4. Interfaces

The interface is where users interact with technology. It must be smart and simple, allowing users to accomplish their goals effectively. Whether built-in or external, interfaces should clearly communicate their affordances and limitations.

5. Context

Context plays a vital role in user interactions. Designers use techniques like contextual inquiry and interviews to observe real

A tip to remember these components with a simple sentence:

Users have **tasks** to complete. Designers use **tools** to create **interfaces** that work effectively in different **contexts.**

EXAMPLE CASE STUDY FOR HCI COMPONENTS: SMART MUSEUM GUIDE

Suppose you are designing an interactive system for a museum.

1. Users

Who will use the system?

Children

Parents

Elderly visitors

Tourists

Researchers

People with disabilities

EXAMPLE: SMART MUSEUM GUIDE

Different users have different needs.

For example:

Children prefer games and animations.

Researchers need detailed historical information.

Elderly visitors require larger fonts.

International tourists need multiple languages.

Should one interface work for everyone?

2. Tasks

What do users want to accomplish?

- Buy tickets
- Find an exhibit
- Learn about an artifact
- Navigate to another gallery
- Listen to an audio guide
- Save favorite exhibits
- Share photos

Example task:

"A tourist wants to find the dinosaur exhibit within two minutes.“
Sample workflow on next slide.

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Example task:

"A tourist wants to find the dinosaur exhibit within two minutes.“
Sample workflow on next slide.

Start

↓

Open Museum App

↓

Search Dinosaur

↓

View Map

↓

Follow Route

↓

Reach Exhibit

3. Tools

Designers use different tools throughout the project.

Design

Figma

Adobe XD

Wireframing (process of creating a basic, skeletal blueprint of a website or app)

Pencil Project

Balsamiq

User Testing

Maze

Google Forms

4. Interface

The interface may include:

Touchscreen kiosk

Mobile app

QR code scanner

Voice assistant

Interactive digital map

5. Context

Visitors use the app in different situations. For Example:

Standing

Walking

Taking photographs

Talking with family

Using one hand

Low battery

No internet

The interface should adapt.

Example:

If the visitor enters the dinosaur gallery,
the app automatically displays dinosaur information.

This is **context-aware computing**.

CLASS EXERCISE:

Divide the class into groups, with each group assigned will take a different real-world scenario. For example: Food Ordering Application, Smart Museum , Smart Zoo, Airport Check-in System, Theme Park, Ride booking App, Smart sports stadium etc.

Each group should imagine that they are part of an HCI design team responsible for developing a digital solution. The goal is to create a user-centered interface that improves the visitor experience.

Every group will have to answer these five questions:

1. Who are the users?
2. What tasks do they want to perform?
3. Which design tools would you use to create the interface?
4. What type of interface would you design (mobile app, kiosk, website, AR/VR, voice assistant, etc.)?
5. What contextual factors (location, time, weather, user type, device, language, activity) should the system consider?

IMPORTANCE OF HCI

Human-Computer Interaction (HCI) is essential for creating intuitive, accessible, and efficient systems that accommodate users with diverse abilities and expertise. It ensures systems are user-friendly, helping users interact with technology without needing to understand its complexities. Here's how HCI impacts our daily lives:

1. **HCI in Daily Life**: HCI is everywhere – from ATMs and vending machines to self-service booths. These systems rely on intuitive interfaces designed to make interaction simple and efficient, regardless of technical expertise.

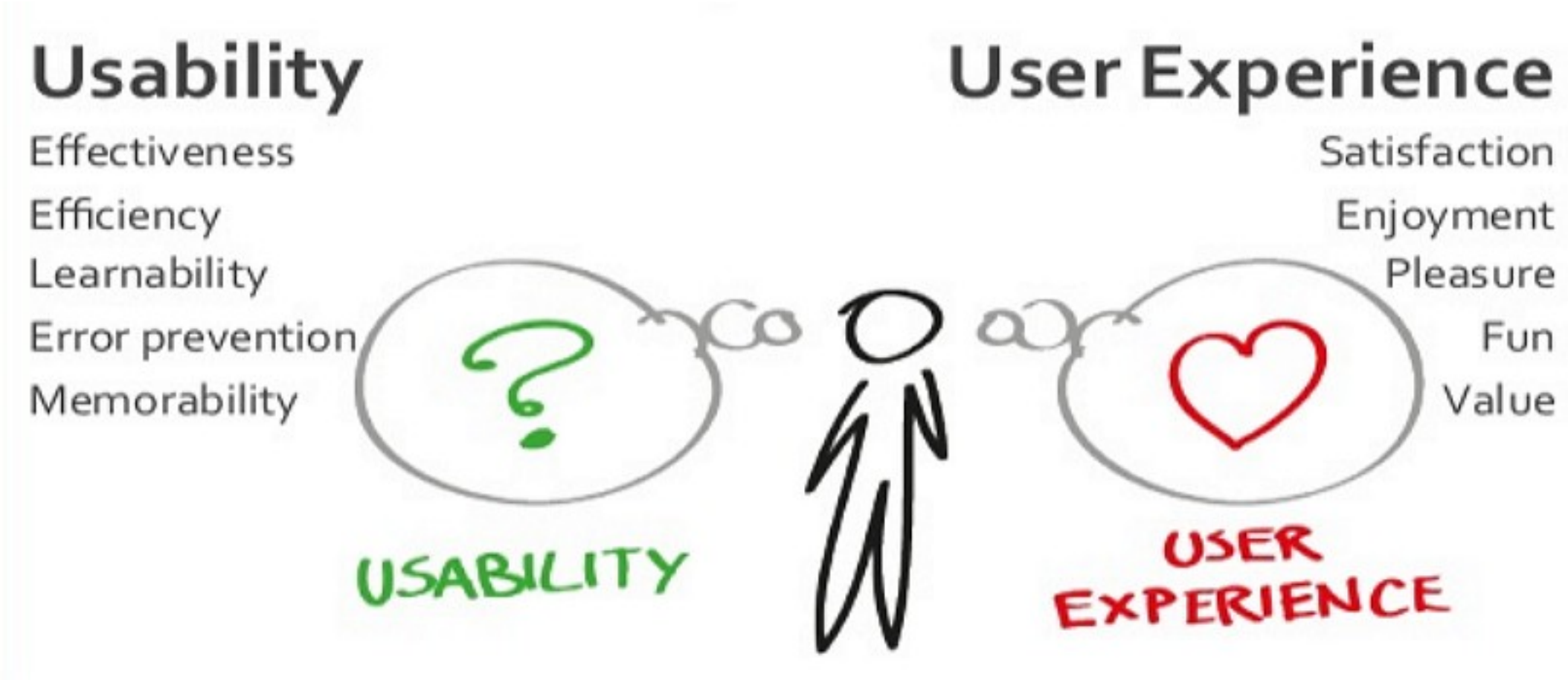
2. **In Industry**: In sectors like air traffic control or power plants, HCI ensures that safety-critical systems are accessible even to non-experts. Well-designed systems boost productivity and ensure safety in high-risk environments.

3. **Accessibility for Disabled**: HCI focuses on making systems usable by people with disabilities, using user-centered design principles to create accessible interfaces, such as voice control and screen readers.

4. **Software Success**: HCI is crucial for the success of software products. By applying usability principles, companies ensure their software is intuitive, which directly impacts user adoption and sales.

5. **For Untrained Communities**: HCI principles enable systems to be intuitive and

HOW TO ACHIEVE EFFECTIVE HCI?



HOW TO ACHIEVE EFFECTIVE HCI?

To achieve effective HCI, developers must focus on **usability** and **user experience**:

1. Usability

Usability ensures users can quickly and easily learn and use systems. A usable system has these characteristics:

- **Learnability**: Easy to learn and remember, especially for new or infrequent users (e.g., user-friendly operating systems like Windows).
- **Safety**: Protects users from errors and dangerous situations, with features like error prevention or recovery (e.g., undo functions or warning messages).
- **Efficiency**: Accomplishes tasks effectively, supporting users in completing their objectives.
- **Effectiveness**: Delivers high-quality performance to achieve desired outcomes.
- **Utility**: Provides the necessary tools and functions for task completion (e.g., IDEs with helpful suggestions).
- **Enjoyability**: The system is pleasant to use, with a simple, intuitive interface

HOW TO ACHIEVE EFFECTIVE HCI?

2. User Experience (UX)

UX focuses on how users feel while interacting with a system. It's subjective and involves:

- **Desirable traits**: Systems that are satisfying, enjoyable, or motivating.
- **Undesirable traits**: Systems that are frustrating, unpleasant, or annoying.

By studying these feelings, developers can refine the system to improve users' emotional responses and create a more positive overall experience.

GOALS OF HCI

The primary goal of HCI is to create usable, safe, and efficient systems for users. Developers can achieve this by:

1. Understanding how users interact with computing systems.
2. Designing tools and methods that cater to users' needs.
3. Continuously testing, refining, and validating the system for effective user interaction.
4. Prioritizing end-users and establishing a strong HCI foundation.

EXAMPLES OF HCI IN TECHNOLOGY

1. **IoT Technology:** IoT devices, like smart thermostats and wearables, collect user data to improve interactions. Pre-touch sensing, where devices predict actions before they are explicitly commanded, is an example of advanced HCI in IoT.

2. **Eye-Tracking Technology:** Eye-tracking systems, used in applications like driver monitoring and employee focus management, detect where users are looking and optimize interactions based on gaze patterns.

3. **Speech Recognition:** Virtual assistants like Alexa, Siri, and Google Assistant interpret and respond to voice commands, enhancing user interaction by allowing hands-free control of devices.

4. **AR/VR Technology:** Augmented Reality (AR) and Virtual Reality (VR) technologies, combined with HCI, provide immersive experiences in applications like training, gaming, and remote work. For example, AR glasses can guide users in real time, adjusting interactions based on their needs.

5. **Cloud Computing:** Cloud-based tools enable remote work, making it easier to collaborate and access data from anywhere. HCI ensures that

HCI CONTRIBUTION TO VARIOUS DISCIPLINES

There is a wide range of topics that HCI covers. The **development of HCI** relies on contributions from different disciplines. Here are some prominent disciplines where HCI is highly considered:

- Cognitive Psychology – Capabilities, information processing, limitations, performance prediction, and cooperative working.
- Human Factors/Ergonomics – display readability and hardware design
- Social Psychology – organizational and social structures
- Linguistics – natural language interfaces
- Computer Science – design, development, and maintenance of software, UIMS (user interface management system), UIDE (user interface development environments), graphics, and prototyping tools.
- Sociology, Anthropology, & Philosophy – computer supported cooperative networks (CSCW)
- Artificial intelligence – Intelligent software
- Engineering & design – engineering principles and graphics